

MERIDIAN BIOSYSTEMS

SBIR Phase I Project Proposal

Project Title: Next-Generation Microfluidic Optoelectronic Biosensor for Rapid Point-of-Care Pathogen Detection

Agency: NSF SBIR Phase I
Topic ID: BT1.2 Biological Tech
Budget: \$275,000
Duration: 6 Months
Date: October 2026

Applicant Firm Details

Company Name: Meridian BioSystems Inc.
DUNS / UEI: K9X8J2M4L7V1
Principal Investigator: Dr. Eleanor Vance, Ph.D.
Location: Cambridge Innovation Center, MA
Contact Email: evance@meridianbio.com
Commercial Track: Medical Diagnostics

Key Performance Metrics

Target Assay Time: < 15 Minutes
Limit of Detection (LOD): < 100 CFU/mL
Sample Matrix: Whole Blood / Plasma
Target Unit Cost: < \$12.50
IP Protection: 2 Provisionals Filed (2025/2026)
Phase II Target: \$500k Match Funding

Executive Summary & Technical Abstract:

Hospital-acquired infections and bloodstream pathogens account for over 270,000 annual fatalities in the United States, largely driven by diagnostic delays of 24–48 hours required for standard microbial blood culture. Meridian BioSystems proposes a novel microfluidic optoelectronic biosensor platform (*OptoDx*) that enables fully automated multiplexed pathogen detection directly from whole blood in under 15 minutes. Phase I research will establish feasibility by optimizing functionalized plasmonic microchannels, achieving sub-picomolar target capture efficiency, and demonstrating benchtop detection of *Staphylococcus aureus* and *Escherichia coli* at < 100 CFU/mL without prior sample enrichment.

1 Identification and Significance of the Innovation

1.1 Market Need and Technical Problem

Current clinical diagnostic protocols for bloodstream infections rely on time-intensive enrichment cultures followed by automated mass spectrometry (MALDI-TOF) or polymerase chain reaction (PCR). While highly specific, these workflow pipelines suffer from three fundamental bottlenecks:

- Diagnostic Delay:** Turnaround times of 24 to 72 hours force clinicians to prescribe broad-spectrum empiric antibiotics, accelerating drug-resistant strain emergence.
- Sample Preparation Complexity:** PCR and nucleic acid amplification methods demand labor-intensive extraction, centrifugation, and skilled laboratory staff.
- Point-of-Care Incompatibility:** Centralized testing infrastructure cannot support decentralization to urgent care clinics, emergency departments, or rural triage centers.

1.2 The Meridian OptoDx Platform Innovation

Meridian BioSystems addresses these limitations by coupling nanostructure-enhanced surface plasmon resonance (SPR) optics with self-driven microfluidic arrays. The core innovation comprises a multi-layered cyclic olefin copolymer (COC) cartridge coated with localized gold nanostar arrays functionalized with high-affinity synthetic aptamers.

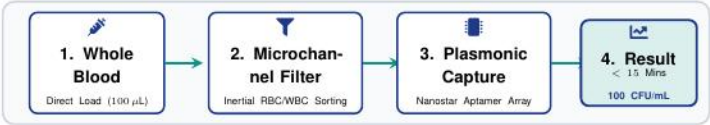


Figure 1: Architectural Overview of the Meridian *OptoDx* Sample-to-Answer Microfluidic Pipeline.